

IIT JAMMU SUMMER SCHOOL 2026

IoT & Drones Program

Assignment: Fundamentals of Drones and UAV Systems

Course IoT & Drones Program	Name _____
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Instructions

Answer all questions neatly and in order. Draw labeled diagrams wherever required. Support your answers with real-world examples and correct technical terminology. Cite all external resources used.

Assignment Questions (100 Marks)

Q1. What is a drone (UAV)? Explain its definition, history, and evolution. *[5 Marks]*

Q2. List at least five real-world applications of drones. *[5 Marks]*

Q3. Differentiate between: (a) UAV and Drone, (b) Fixed-wing and Multirotor drones, (c) Quadcopter and Hexacopter. *[10 Marks]*

Q4. Explain how a quadcopter flies. Include lift, thrust, gravity, drag, propeller rotation, and clockwise (CW) / counter-clockwise (CCW) motors. Draw a neat diagram. *[10 Marks]*

Q5. Explain how a drone performs take-off, landing, hovering, move forward, move backward, move left, move right, and rotate (yaw). Use suitable diagrams. *[10 Marks]*

Q6. Draw a labeled block diagram of a drone and explain the function of each component: flight controller, ESC, brushless motors, propellers, battery, power distribution board, GPS module, radio receiver, frame, telemetry module. *[10 Marks]*

Q7. Explain the role of the following components: flight controller, ESC, BLDC motor, propeller, Li-Po battery, GPS, radio transmitter, frame, battery eliminator circuit (BEC), telemetry module. *[10 Marks]*

Q8. Explain the working principle and application of IMU, accelerometer, gyroscope, magnetometer, barometer, GPS, optical flow sensor, ultrasonic sensor, LiDAR, and camera. For each sensor mention purpose, working principle, and application. *[10 Marks]*

Q9. Compare: (a) GPS vs Optical Flow, (b) LiDAR vs Ultrasonic, (c) Gyroscope vs Accelerometer. *[5 Marks]*

Q10. Explain the concept of 6 Degrees of Freedom (6DOF). Include translation, rotation, X-axis, Y-axis, Z-axis. Draw suitable diagrams. *[5 Marks]*

Q11. Explain roll, pitch, yaw, surge, sway, and heave with examples of each movement in a drone. *[10 Marks]*

Q12. Explain the complete signal flow in a drone: Pilot Input to Radio Transmitter to Receiver to Flight Controller to ESC to BLDC Motors to Propellers to Drone Movement. Illustrate with a block diagram. *[5 Marks]*

Q13. Why is PID control important in drones? Explain the proportional, integral, and derivative terms.
[5 Marks]

Q14. Bonus — Research Questions. Answer any TWO of the following: [10 Marks]

- Compare DJI, ArduPilot, and PX4 flight controllers.
- Explain obstacle avoidance in drones.
- How do drones return to home automatically?
- Explain drone swarm technology.
- What is autonomous navigation?
- Explain computer vision in drones.
- Discuss future trends in drone technology.

Q15. Practical Activity — Choose ONE: [10 Marks]

- Identify and label every component of a real drone or a drone image.
- Create a 3D CAD model or hand sketch of a quadcopter.
- Use a drone simulator and document take-off, hover, roll, pitch, yaw, and landing with screenshots/video and observations.

Higher-Order Thinking Questions

Reflect on and answer the following in your own words: (NO AI Based Answers)

- **1.** Why do quadcopters require four motors instead of one?
- **2.** What would happen if one motor stops working during flight?
- **3.** Why are brushless motors preferred over brushed motors?
- **4.** Can a drone fly indoors without GPS? Explain.
- **5.** Why are Li-Po batteries used in drones instead of standard batteries?
- **6.** What factors reduce a drone's flight time?
- **7.** How can drone stability be improved during strong winds?
- **8.** Why is sensor fusion important in drones?
- **9.** Explain how GPS and IMU work together for stable navigation.
- **10.** If you were designing a drone for mountain rescue operations, which sensors and components would you choose, and why?

Submission Requirements

- Cover page with name, course, and date.
- Answers with clear, labeled diagrams and figures wherever required.
- Proper citations for all external resources used.
- Clear figures and tables, neatly presented.